

Different Quantum Computers

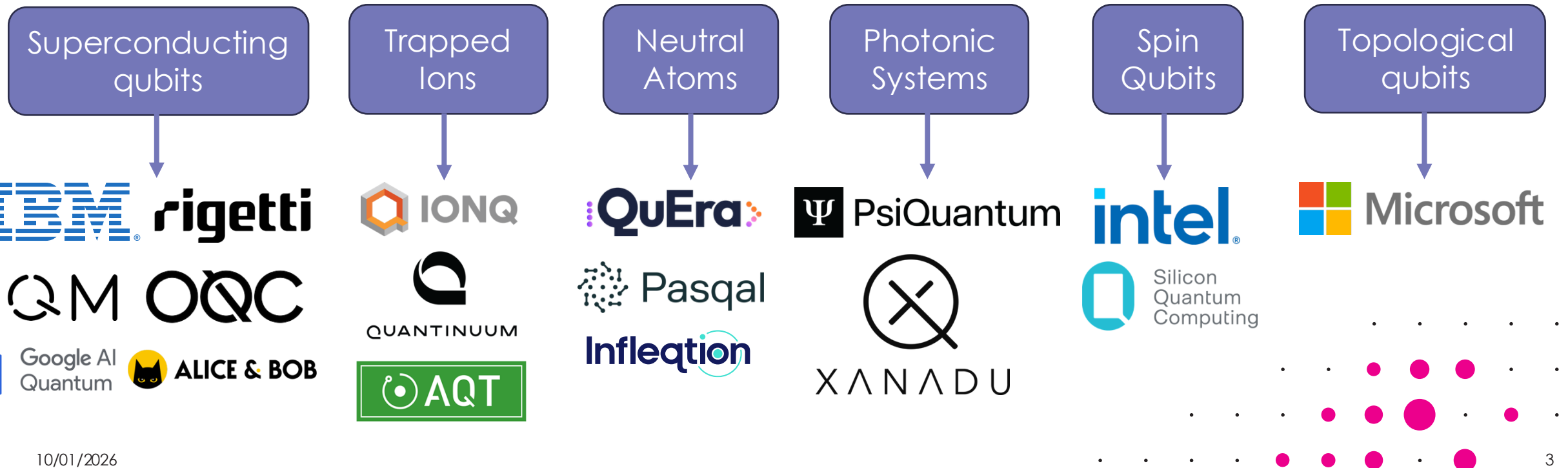
Introduction to Quantum Computing course 2026 – Meeri Mäkitalo

Outline

1. The quantum hardware landscape
2. Main qubit architectures
 1. **Superconducting qubits**
 2. Trapped ions
 3. Neutral atoms
 4. Photonic quantum computers
 5. Spin qubits
 6. Topological qubits
3. Gate-based systems vs. Quantum annealers

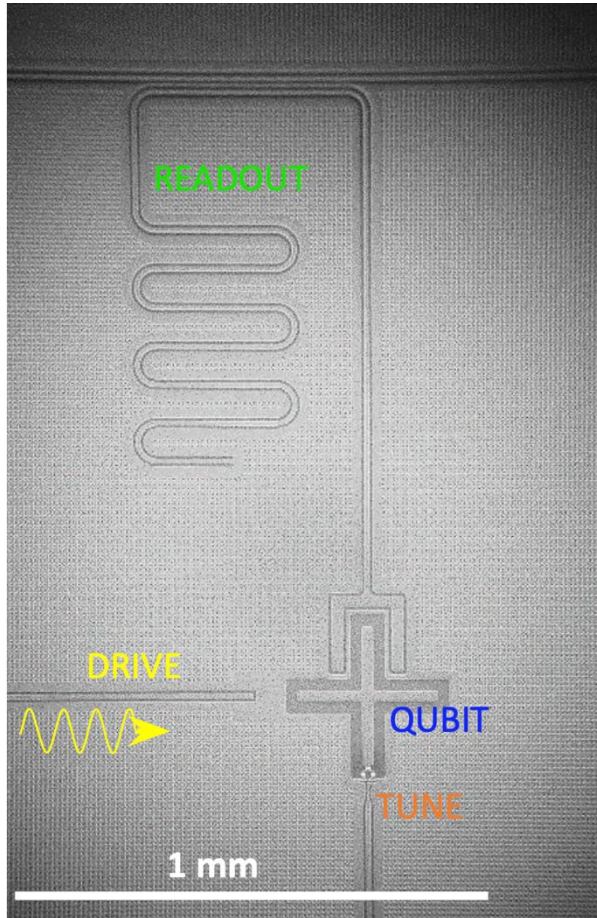
The quantum hardware landscape

- Not one unique way to make qubits
 - Classical computers use silicon transistors
 - How to make the best qubit has not been defined yet
- Goal: a quantum system that can be isolated from the environment and controlled, measured, and scaled easily

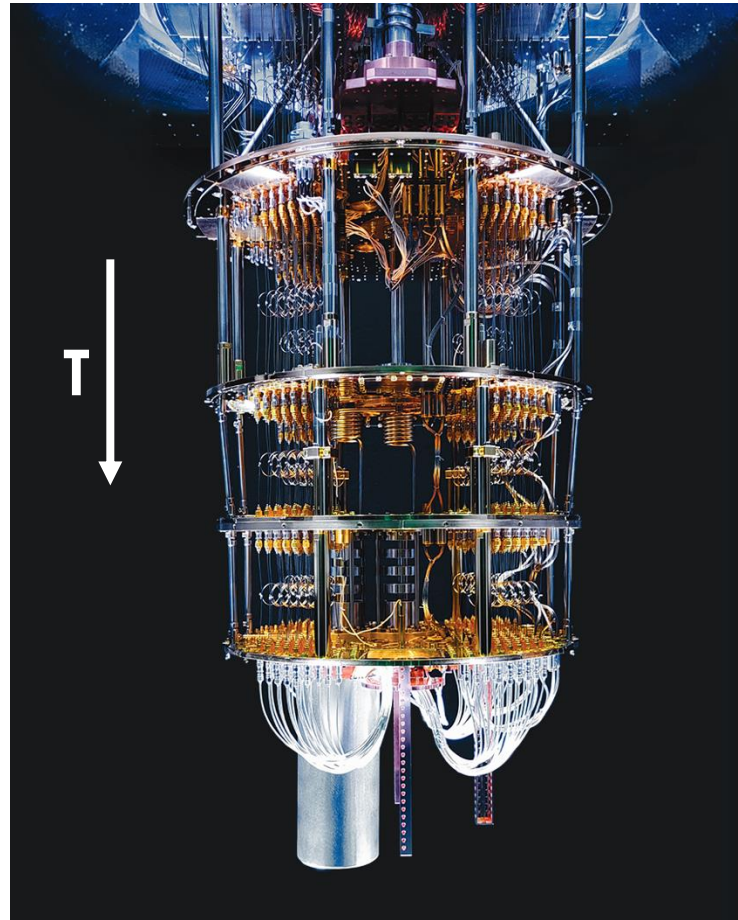


Superconducting quantum computers

Circuit printed on a (silicon) chip



Dilution refrigerator



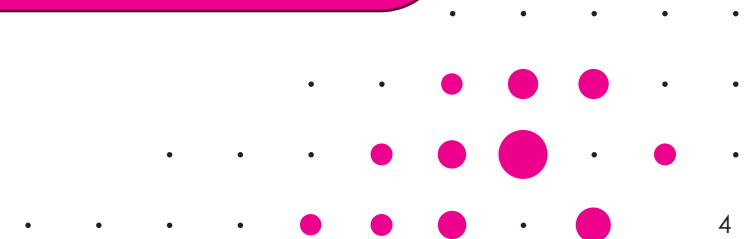
Pros:

- Fast gate speeds
- Similar fabrication to classical silicon chips
- Tunability
- Strong coupling

Cons:

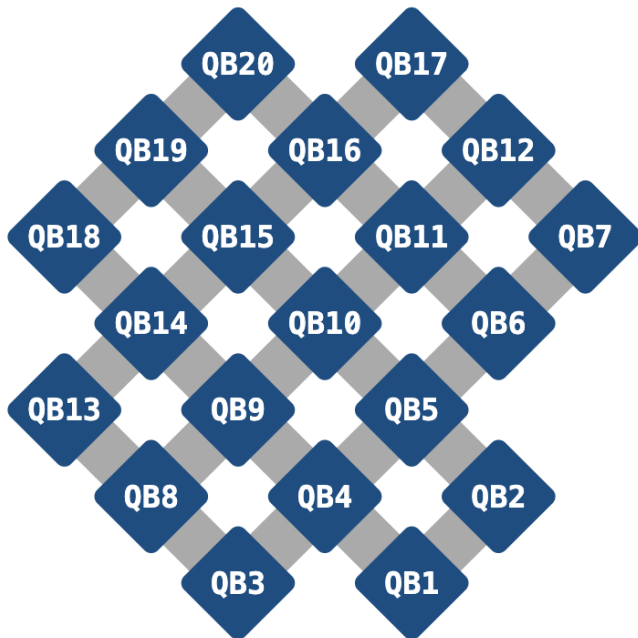
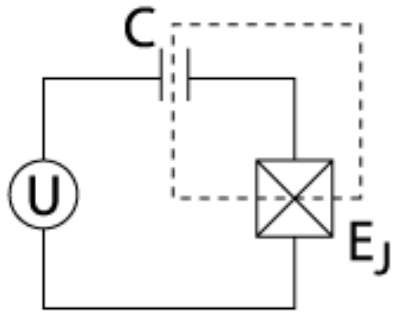
- Prone to environmental noise
- Need for calibration
- Wiring bottleneck

image credit: Suman Kundu, Aalto University and IQM



Superconducting qubits

A charge qubit $\longrightarrow$ A transmon qubit



The Josephson junction

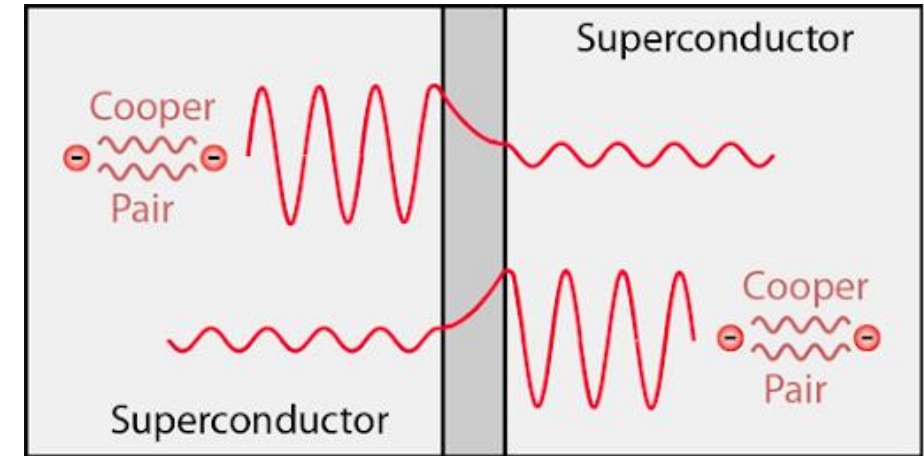
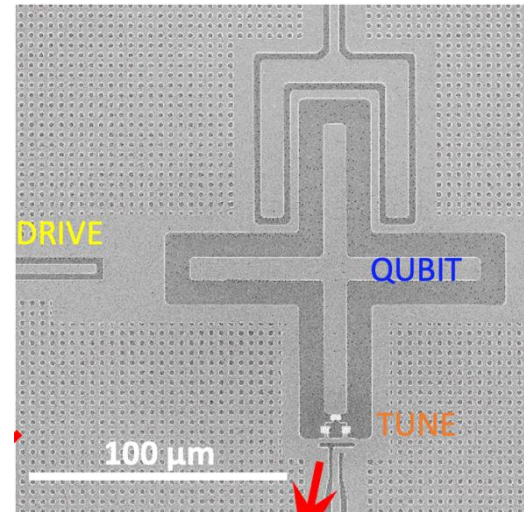


image credit: Integrated Silicon-Metal Systems At the Nanoscale, Applications in Photonics, Quantum Computing, Networking, and Internet

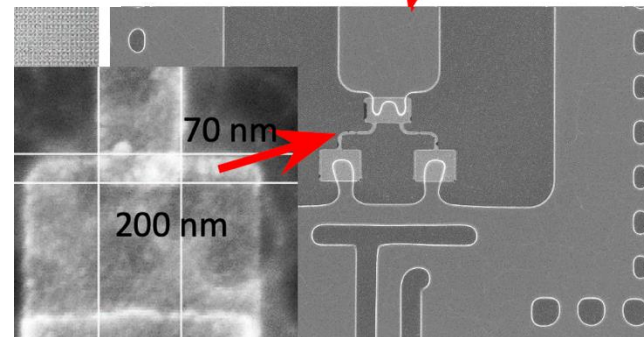
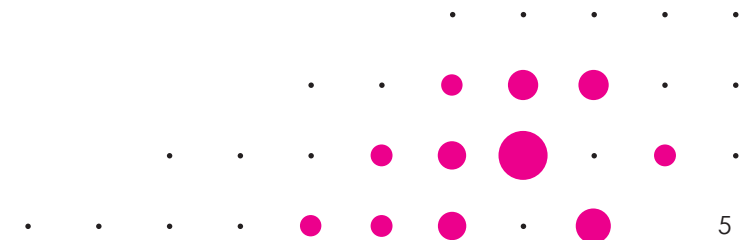


image credit: Suman Kundu, Aalto University



Trapped ion quantum computers

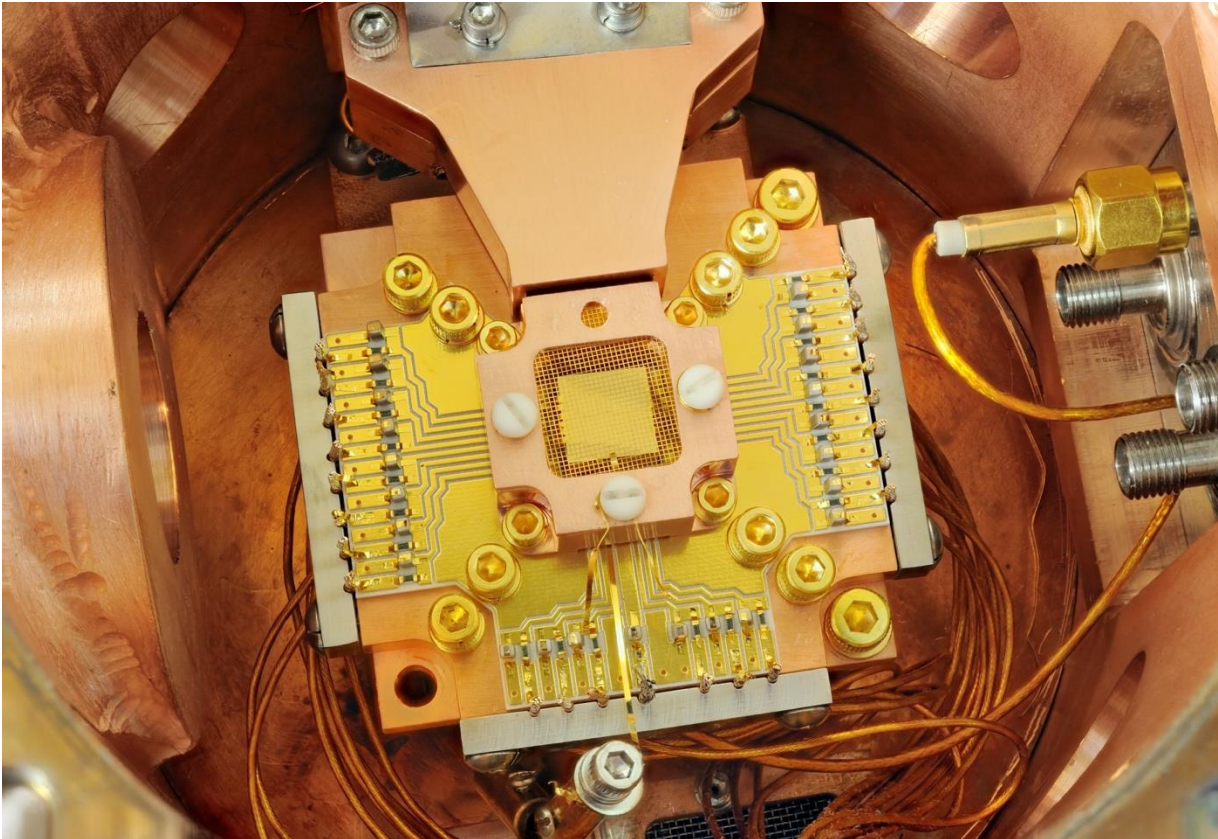


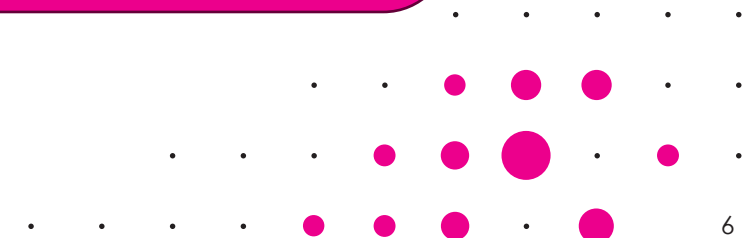
Image credit: National Institute of Standards and Technology

Pros:

- Very stable
- Identical qubits
- High fidelities
- All to all connectivity

Cons:

- Slow operation speeds
- Setup is bulky
- Scaling is hard



Neutral atom quantum computers

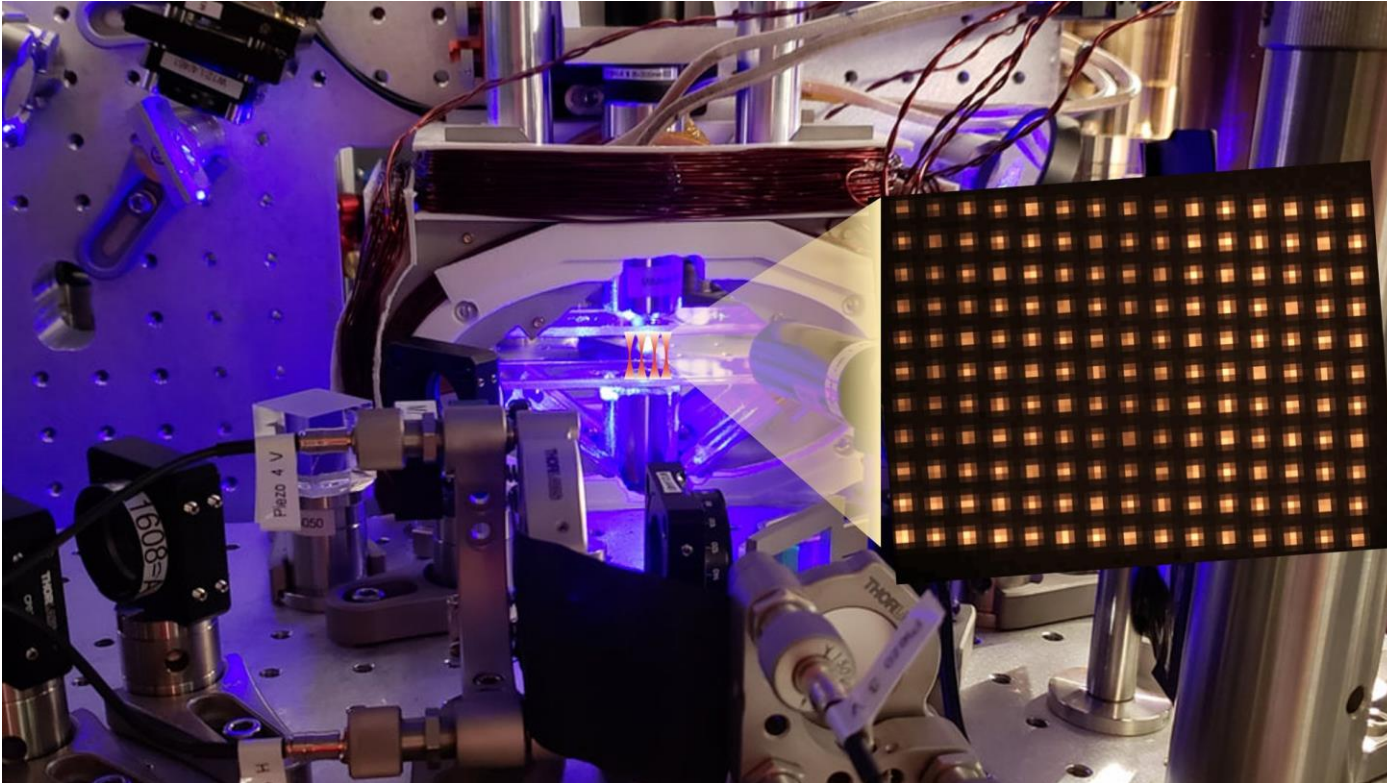


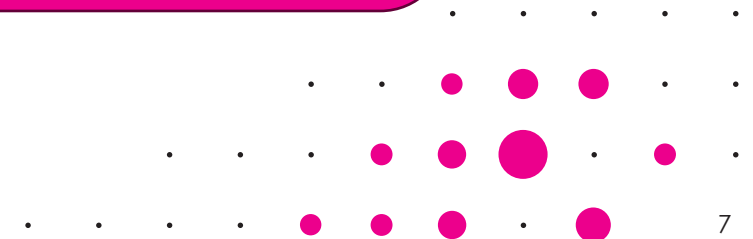
Image credit: Choi research group Stanford University

Pros:

- Identical qubits
- Vacuum chamber at room temperature
- Can be tightly packed

Cons:

- Sensitive to environmental noise
- Sustaining precise laser power



Photonic quantum computers

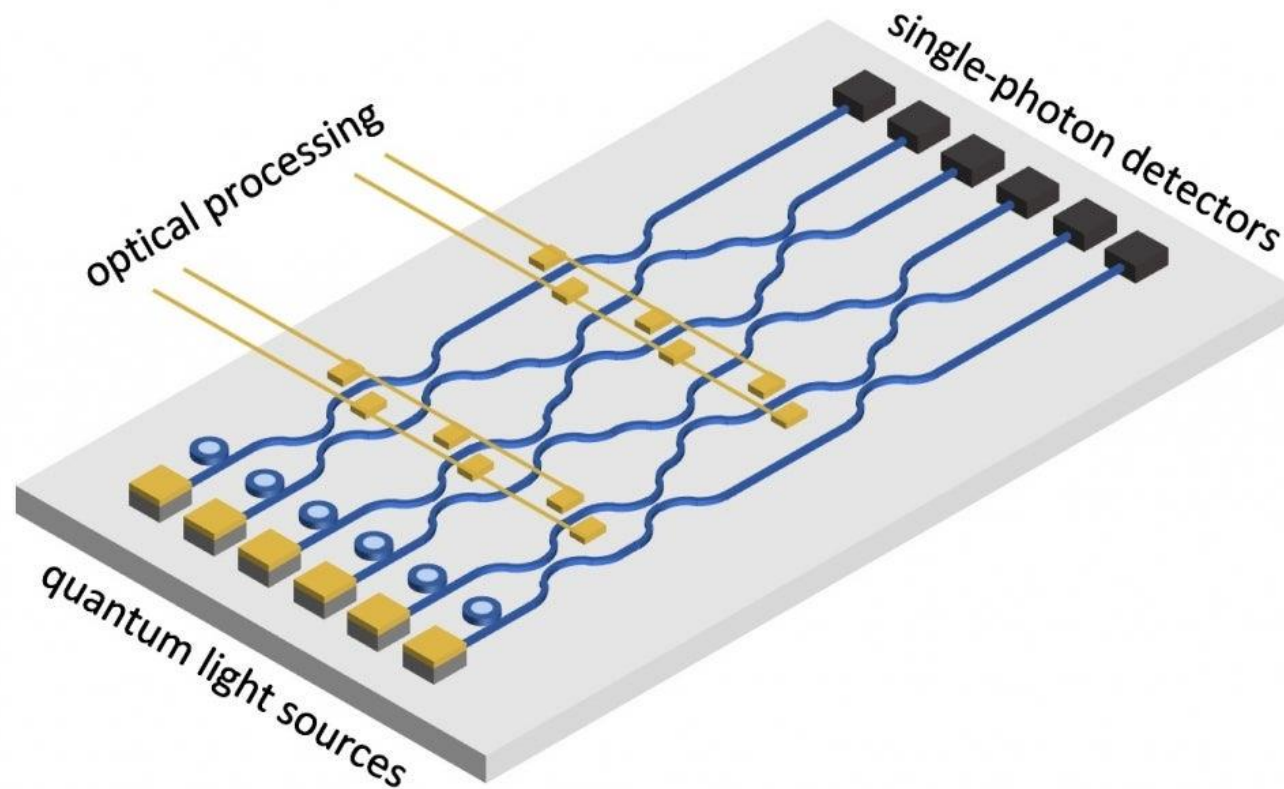


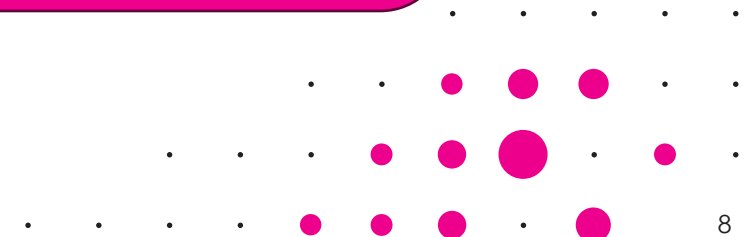
Image credit: UC Santa Barbara

Pros:

- Long coherence times
- Mostly in room temperature
- Modular and easily integrated

Cons:

- Performing multi qubit gates is difficult
- "Photon loss"



Spin qubits

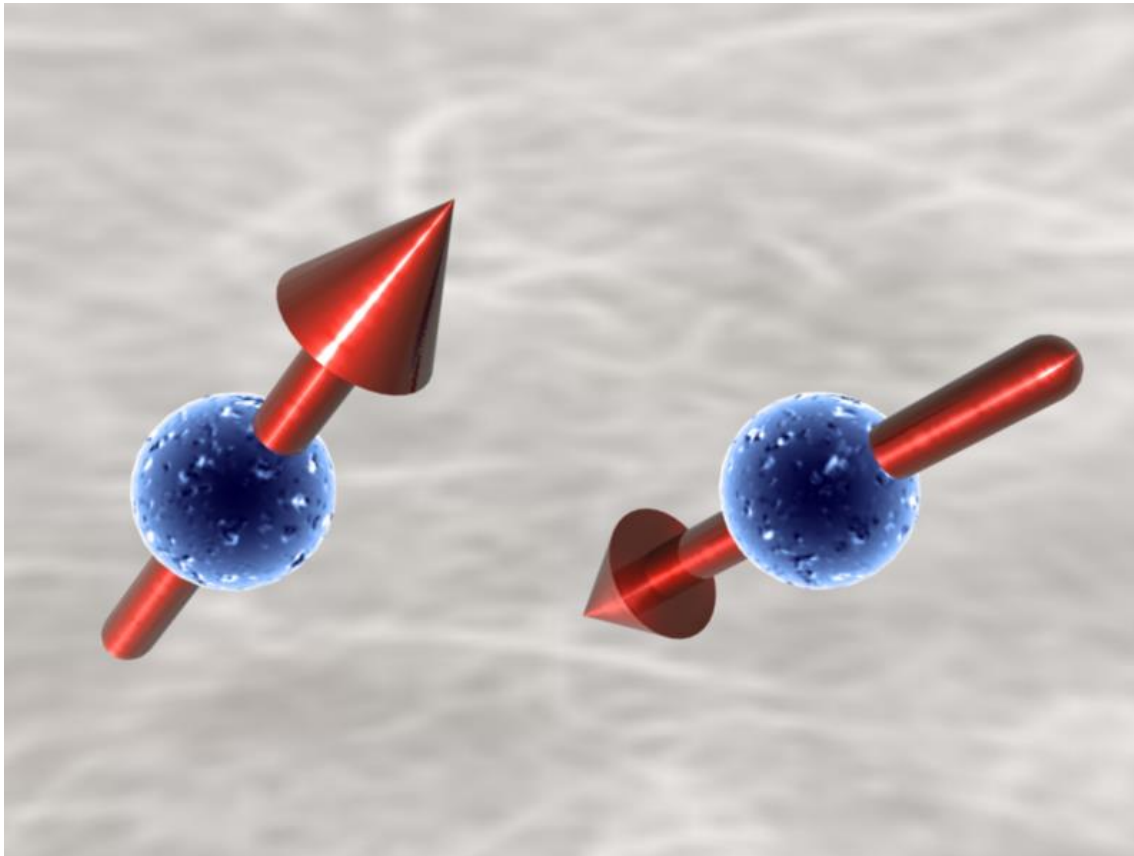


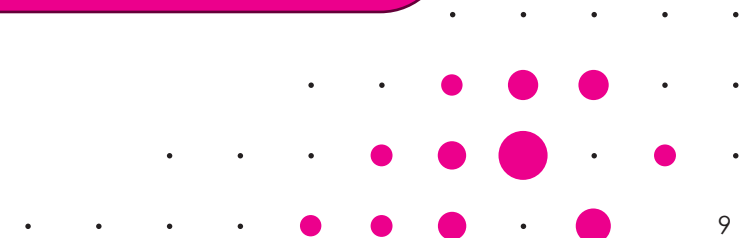
Image credit: QuTech

Pros:

- Very small
- Leverage existing semiconductor technology
- Operate at slightly warmer T (~1K)

Cons:

- Sensitive to noise
- Tight packing → interference



Topological qubits

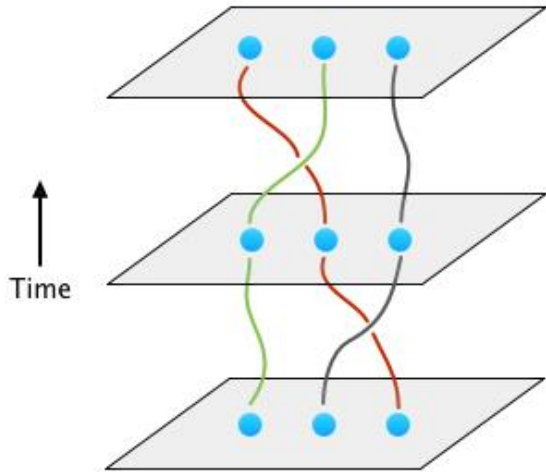
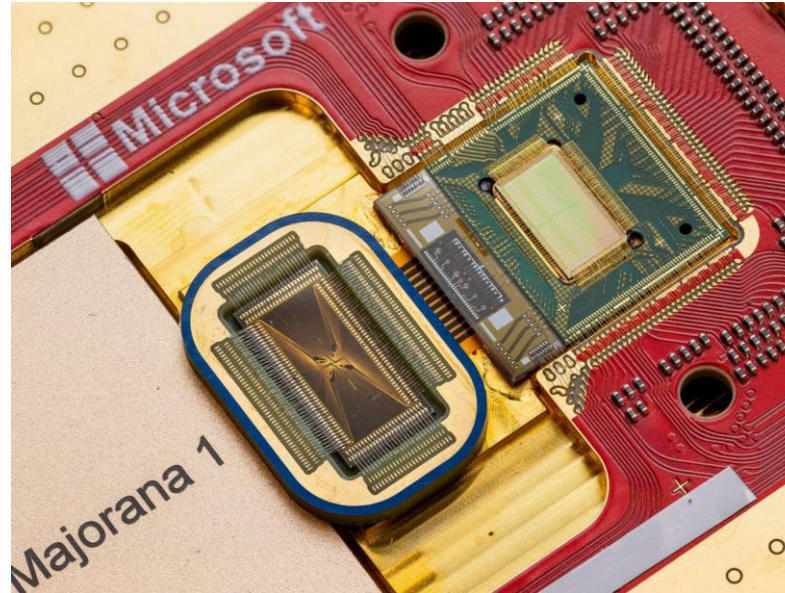


Image credits: Quanta Magazine and Microsoft



Pros:

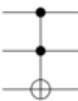
- Noise resistant
- Hardware-level error correction

Cons:

- Least mature of the options
- Isolation and controlling remains a challenge

NOKIA

Gate based systems vs. Quantum annealers

Operator	Gate(s)	Matrix
Pauli-X (X)		$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$
Pauli-Y (Y)		$\begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}$
Pauli-Z (Z)		$\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$
Hadamard (H)		$\frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$
Phase (S, P)		$\begin{bmatrix} 1 & 0 \\ 0 & i \end{bmatrix}$
$\pi/8$ (T)		$\begin{bmatrix} 1 & 0 \\ 0 & e^{i\pi/4} \end{bmatrix}$
Controlled Not (CNOT, CX)		$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$
Controlled Z (CZ)		$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \end{bmatrix}$
SWAP		$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$
Toffoli (CCNOT, CCX, TOFF)		$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}$

High Energy

Low Energy

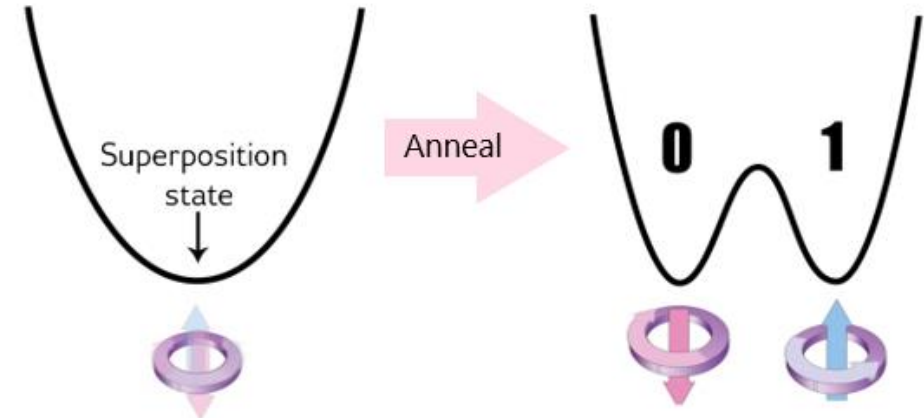
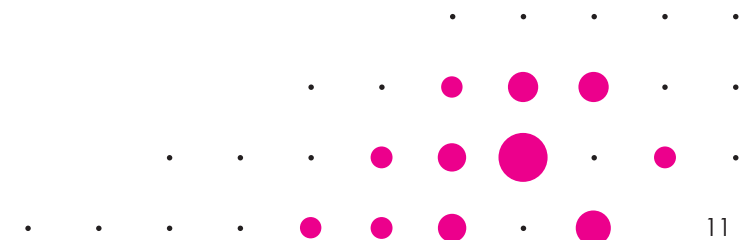


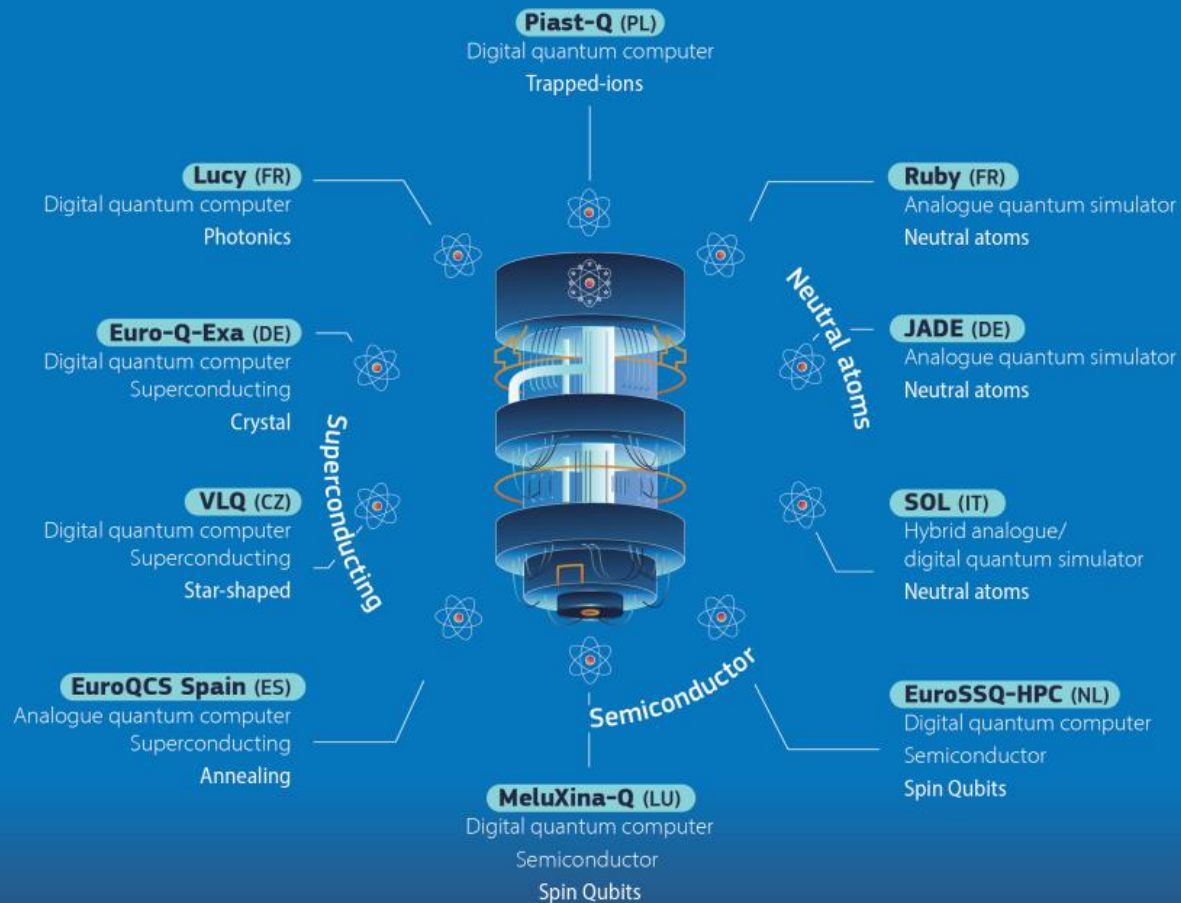
Image credit: D-Wave

D:wave
The Quantum Computing Company™





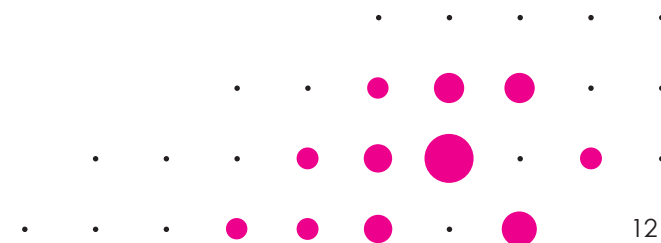
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Application Specialist in the Quantum Technologies team at CSC

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